Required by the syringe's filling step: **BottomUpFilling**, 1.0 mL;
the needle at most 2 mm above, 3 mm below the liquid;
the barrel 6.35 mm wide, its bottom 11 mm above the nest

Task from the vocabulary: BottomUpFilling ⊑ Filling

pre: empty | post: content 1.0 mL | keep: −2 ≤ immersion ≤ 3

Actions from the Capability Description

Dosing: content += V at a rate r, 0.1 to 5 mL/s;
immersion = y − (70 − 11) + 31.6 · content

Positioning: the needle to y at 12.5 mm/s; y ≤ 70 − 11 − 1

Goal regression: content = 1.0 mL

Dosing alone at 40 mm:
immersion −19 → +12.6

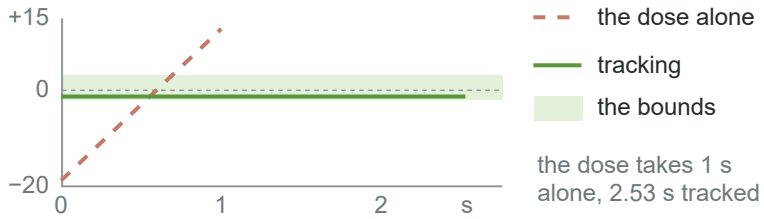
broken at its start and its end, so not applied: the next method

Tracking: Dosing ∥ Positioning

- 31.6 · r = 12.5, so r = 0.396 mL/s
- starts at y_0 in [57, 58]: 57.5
- rises 31.6 mm in 2.53 s, to 25.9

In time: how deep the needle is in the liquid

mm, + in the liquid



the dose takes 1 s alone, 2.53 s tracked

The planned skill: the pair a parallel block

MoveAxis 57.5 → (Dispense 1.0 at 0.396 mL/s ∥ MoveAxis 25.9) → Home

Checked in time, described:
Skills, *Described*

Offered BottomUpFilling
after a reconfiguration